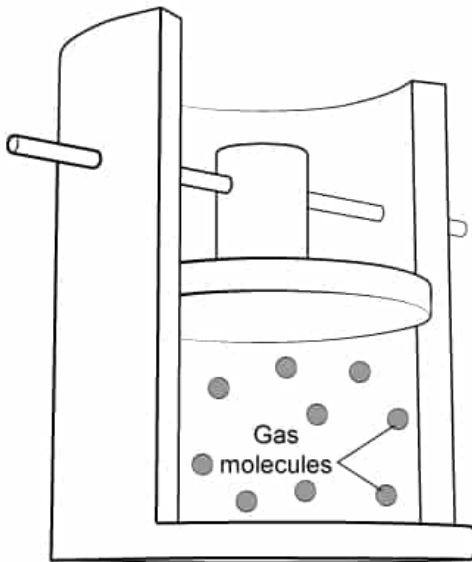


A monoatomic ideal gas is initially sealed within a container featuring a piston fixed in place at one end.



The piston is then unlocked and allowed to move under constant pressure. How does the initial heat capacity of the gas at constant volume C_V compare with the heat capacity of the gas at constant pressure C_P and why?

- ✔ ☒ A. C_P will be greater than C_V , because the gas can expend energy by doing work on the piston. [48%]
- ☐ B. C_P will be less than C_V , because the movement of the piston will prevent heat transfer. [7%]
- ☐ C. C_P will equal C_V , because the internal energy of the gas can no longer change. [12%]
- ☐ D. C_P will equal C_V , because the gas molecules are monoatomic and ideal. [31%]

Correct

48% answered correctly

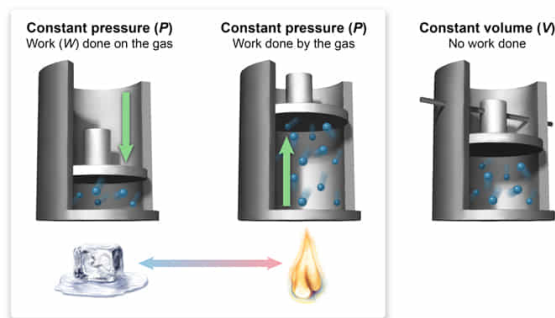
Explanation:

Molar heat capacity at constant pressure & volume

$$\Delta U_P = (Q - W) = (nC_P \Delta T - P \Delta V)$$

$$\Delta U_V = Q = nC_V \Delta T$$

- $\Delta U_P, \Delta U_V$ Change in internal energy (constant P, V)
- $\Delta C_P, C_V$ Molar heat capacity (constant P, V)
- ΔT Change in temperature
- Q Heat energy
- n Moles of gas



Molar heat capacity at constant pressure (C_P) exceeds molar heat capacity at constant volume (C_V) because heat (Q) is lost as work

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The **molar heat capacity (C)** is the amount of heat required to raise the temperature of 1 mole of a substance by 1 degree Kelvin. As such, the **heat (Q)** associated with **n moles** of a substance undergoing a temperature change (ΔT) is given by:

$$Q = nC \Delta T$$

Because the internal energy (U) of a **monoatomic ideal gas** depends on the **gas temperature**, the molar heat capacity at constant volume (C_V) is not equal to the molar heat capacity at constant pressure (C_P).

At **constant volume (V)**, the change in internal energy (ΔU) of the gas is equal to the heat energy:

$$\Delta U_V = Q = nC_V \Delta T$$

However, at **constant pressure (P)**, ΔU depends on both heat and work (W), because the **ideal gas law** states that a temperature change from a heat transfer must produce a change in volume (ΔV) under constant pressure:

$$\Delta U_P = (Q - W) = (Q - P \Delta V)$$

$$\Delta U_P = nC_P \Delta T - P \Delta V$$

If the internal energy change (ie, temperature change) is assumed to be constant between constant volume and constant pressure conditions, the expression for ΔU_V can be substituted into the equation above to give:

$$\Delta U_V = \Delta U_P$$

$$nC_V \Delta T = nC_P \Delta T - P \Delta V$$

$$C_V = C_P - \frac{P \Delta V}{n \Delta T}$$

Moreover, because the ideal gas constant R is equal to $\frac{P \Delta V}{n \Delta T}$, the equation simplifies to:

$$C_V = C_P - R$$

This expression shows that C_V is less than C_P because some heat energy transferred into a gas stored at constant pressure may be converted to pressure-volume work during the expansion or compression of the gas.

Therefore, allowing a previously fixed piston to move freely (to maintain constant pressure) will increase the molar heat capacity of a monoatomic ideal gas.

(Choices B, C, and D) Heat may transfer into or out of the gas and change the temperature and internal energy of the gas whether the piston is allowed to move or not. However, once the piston becomes mobile, work can occur. As a result, the heat capacity cannot remain constant but will *increase*.

Educational objective:

The molar heat capacity of a gas is the amount of heat required to raise the temperature of 1 mole of the gas by 1 degree Kelvin. The molar heat capacity of a gas at constant pressure exceeds the molar heat capacity of a gas at constant volume.

Time spent:0 sec

Subject:

Physics

QID:402480

System:

Thermodynamics